



NAUVAL ZULFIKAR

Applied ML / NLP Engineer · Responsible AI candidate Bandung, Indonesia (relocatable to Tel Aviv, 2026-10-01) zulfiyar.nauval1998@gmail.com · +44 7300 469048 · +62 821 2567 2264 [LinkedIn](#) · [GitHub](#) · [Portfolio](#) **Visa status:** UK Graduate Visa (active). Indonesian passport. Eligible for EU Researcher (Hosting Agreement) visa for European PhDs.

PROFILE

MSc Business Analytics graduate with a First Class degree from Aston University, specialising in NLP and ML applications within supply chain and governance contexts. My dissertation, supervised by Dr. Ammar Al-Bazi, focused on enhancing supply chain systems through blockchain and LLM analysis, laying a strong foundation for research in AI and NLP. I aim to contribute to the field of responsible AI, particularly in natural language processing and deep learning, aligning with the research themes at Tel Aviv University.

EDUCATION

Aston University, Birmingham, UK (*MSc Business Analytics, 1:1 First Class · EQF Level 7 · Sep 2023 – Sep 2024*)

- **1st Winner, Aston Enterprise Scholarship 2023** (full-tuition competitive award)
- **Aston Pitching Contest 2024** runner-up (AI-powered supply chain proposal)
- **Distinctions:** Decision Models, Effective Management Consultancy, Software Analytics
- **Dissertation:** *Enhancing Supply Chain Information Sharing Systems Through Blockchain: Integrated Customer Reviews LLM Analysis: A Smart Contract Approach* (supervised by **Dr. Ammar Al-Bazi**)

Ritsumeikan Asia Pacific University, Beppu, Japan (*BBA Marketing · EQF Level 6 · Oct 2016 – Sep 2019*)

- APU 50% Tuition Reduction Scholarship + JASSO Scholarship
- **Distinctions:** Business Data Analysis, Consumer Behaviour, Marketing Research

RESEARCH-RELEVANT PROJECTS

LLM-Generated Adaptive Shipper Decision Rules (2026)

- Developed an end-to-end RAG system using gpt-4o-mini for adaptive decision-making in shipping, integrating behavioural literature. [GitHub](#)

Mini Digital Twin for Integrated Passenger-Freight Transport (2026)

- Created a three-layer Digital Twin prototype with live dashboards and interactive simulations. [GitHub](#)

VRPOD MILP Optimisation (2026)

- Implemented MILP optimisation for vehicle routing, achieving a 38.5% cost reduction on a Birmingham-inspired instance. [GitHub](#)

Crowd-Shipping ABM Behavioural Sweep (2026)

- Conducted a 450-run parameter sweep to analyse behavioural responses in crowd-shipping scenarios. [GitHub](#)

INDUSTRY EXPERIENCE

Public Works & Spatial Planning Department (PUTR), Bandung (*Data Scientist & Project Manager · Sep 2022 – Present*)

- Automated ETL pipelines and dashboards, achieving 45% faster project delivery and 20% data-efficiency gain.

Syncwell, Birmingham, UK (*Marketing Data Scientist · Jul – Sep 2024*)

- Implemented predictive ML strategies, resulting in a 40% increase in CTR and 22% follower growth.

PCOS Challenge, Birmingham, UK (*Strategic Data Scientist Specialist · Jun – Aug 2024*)

- Developed customer attribution models and automated reporting, reducing manual workload by 80%.

Bank Muamalat Indonesia, Jakarta (*Lead Data Scientist · Nov 2019 – Aug 2022*)

- Developed NLP models for credit analysis, achieving a 5.3% investment return and maintaining model reliability.

PUBLICATIONS

Zulfikar, N. (2025) *Performance Analysis in Sport Footwear Sales Prediction Using Machine Learning*. Working paper on comparative ML regressor evaluation for retail forecasting.

TECHNICAL SKILLS

ML/NLP: PyTorch, HuggingFace Transformers, DeBERTa-v3 fine-tuning, gpt-4o-mini API, RAG, sentence-transformers

Languages/Infrastructure: Python, SQL, R, PySpark, FastAPI, Streamlit

Mathematical/Formal: MILP, Agent-Based Modelling, Discrete-Event Simulation, Bayesian methods

Governance: Model validation, reliability, audit-grade documentation, AWS, Docker, Git

LANGUAGES

Indonesian: Native

English: Fluent (IELTS 7.0 overall)

Japanese: Intermediate

REFEREES

Dr. Ammar Al-Bazi

Senior Lecturer, Aston Business School, Aston University

a.al-bazi@aston.ac.uk

Dr. Viktor Pekar

Lecturer, Aston Business & Social Sciences

v.pekar@aston.ac.uk